

Regression Estimation in Sampling

Sampling Theory and Methods

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Learning Objectives

By the end of this section, students should be able to:

- 1 Explain the idea of regression estimation in sampling.
- 2 Identify the role of an auxiliary variable in improving the precision of an estimator.
- 3 Derive and interpret the regression estimator of a population mean.
- 4 Estimate the regression coefficient from sample data.
- 5 Obtain the variance of the regression estimator.
- 6 Explain the minimum-variance property of the regression estimator.
- 7 Apply regression estimation in stratified sampling.

Why Regression Estimation?

Basic Idea

Regression estimation uses information from an **auxiliary variable** that is correlated with the study variable.

Suppose:

Y = study variable

and

X = auxiliary variable.

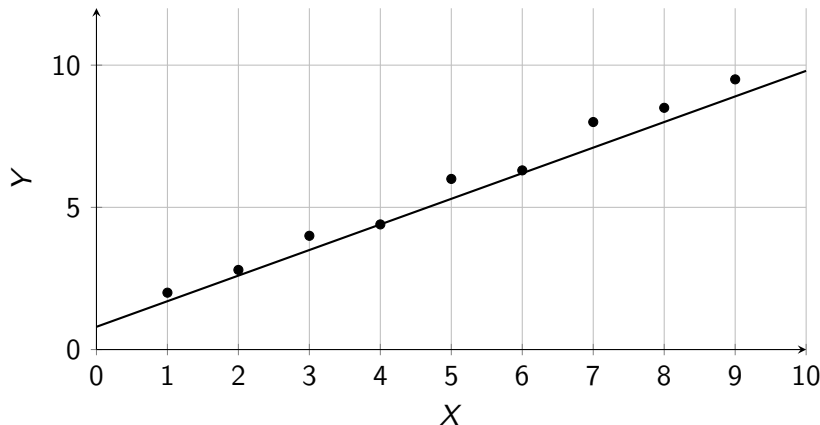
If X is strongly correlated with Y , information about X can be used to improve the estimation of Y .

Key Principle

Use information that is already available about X to obtain a more precise estimate of the population characteristic of Y .

From Ratio Estimation to Regression Estimation

Like a ratio estimator, the regression estimator uses an auxiliary variable X correlated with the study variable Y . However, the relationship between Y and X may be approximately linear without passing through the origin.



Linear Regression Relationship

Assume that the relationship between Y and X can be approximately represented by

$$Y_i = \alpha + \beta X_i + e_i,$$

where

- α = intercept,
- β = regression coefficient,
- e_i = random error.

The regression coefficient represents the expected change in Y associated with a one-unit increase in X .

Sampling Perspective

In sampling, the population regression coefficient is generally unknown and is estimated from the sample.

Notation

Consider a finite population of size N .

$$U = \{1, 2, \dots, N\}$$

A sample of size n is selected.

For each sampled unit: $(y_i, x_i), \quad i = 1, 2, \dots, n.$

Define the sample means:

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

and

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

The population mean of the auxiliary variable is assumed to be known:

$$\bar{X} = \frac{1}{N} \sum_{i=1}^N X_i.$$

Regression Estimator of Population Mean

The population mean of the study variable is

$$\bar{Y} = \frac{1}{N} \sum_{i=1}^N Y_i.$$

The regression estimator is

Regression Estimator

$$\bar{y}_{lr} = \bar{y} + b(\bar{X} - \bar{x})$$

where

$$b = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}.$$

Here, b is the estimated regression coefficient of Y on X .

Interpretation of the Regression Estimator

The estimator

$$\bar{y}_{lr} = \bar{y} + b(\bar{X} - \bar{x})$$

can be interpreted as

Regression estimate = Sample mean + Adjustment

where the adjustment is

$$b(\bar{X} - \bar{x}).$$

- If $\bar{x} = \bar{X}$, then

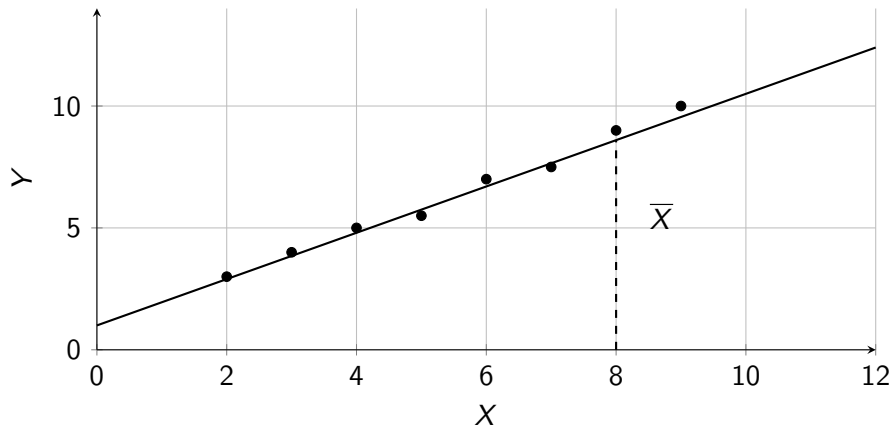
$$\bar{y}_{lr} = \bar{y}.$$

- If $\bar{x} < \bar{X}$ and $b > 0$, the estimate is adjusted upward.
- If $\bar{x} > \bar{X}$ and $b > 0$, the estimate is adjusted downward.

Graphical Interpretation

The regression estimator predicts the population mean of Y at the known population mean $\bar{X}$.

$$\hat{Y}(\bar{X}) = \bar{y} + b(\bar{X} - \bar{x})$$



Sample Regression Coefficient

The estimated regression coefficient is

$$b = \frac{S_{xy}}{S_{xx}}$$

where

$$S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

and

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2.$$

Therefore,

$$b = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}.$$

Interpretation

b measures the estimated change in Y for a one-unit increase in X .

Numerical Example

Suppose a sample of five households provides the following data:

Household	X	Y
1	2	4
2	4	5
3	6	8
4	8	9
5	10	12

Suppose the known population mean of X is

$$\bar{X} = 7.$$

The sample means are

$$\bar{x} = 6, \quad \bar{y} = 7.6.$$

We first calculate the sample regression coefficient b .

Numerical Example: Estimating b

For the data,

$$b = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}.$$

The required quantities are

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = 20,$$

and

$$\sum (x_i - \bar{x})^2 = 40.$$

Therefore,

$$b = \frac{20}{40} = 0.5.$$

Thus the estimated regression coefficient is

Numerical Example: Regression Estimate

The regression estimator is

$$\bar{y}_{lr} = \bar{y} + b(\bar{X} - \bar{x}).$$

Substituting

$$\bar{y} = 7.6, \quad b = 0.5, \quad \bar{X} = 7, \quad \bar{x} = 6,$$

we obtain

$$\begin{aligned}\bar{y}_{lr} &= 7.6 + 0.5(7 - 6) \\ &= 7.6 + 0.5 \\ &= \boxed{8.1}.\end{aligned}$$

Interpretation

Using the known population mean of the auxiliary variable, the estimated population mean of

Variance of the Regression Estimator

Under simple random sampling, the variance can be expressed in terms of the regression relationship.

For a fixed regression coefficient b_0 ,

$$V(\bar{y}_{lr}) \approx \left(1 - \frac{n}{N}\right) \frac{S_y^2 + b_0^2 S_x^2 - 2b_0 S_{xy}}{n}$$

where

S_y^2 = population variance of Y ,

S_x^2 = population variance of X ,

and

S_{xy} = population covariance of X and Y .

Minimum Variance of the Regression Estimator

Consider

$$V(\bar{y}_{lr}) \propto S_y^2 + b_0^2 S_x^2 - 2b_0 S_{xy}.$$

Differentiate with respect to b_0 :

$$\frac{dV}{db_0} = 2b_0 S_x^2 - 2S_{xy}.$$

For minimum variance,

$$2b_0 S_x^2 - 2S_{xy} = 0.$$

Hence,

$$b_0 = \frac{S_{xy}}{S_x^2}$$

which is the population regression coefficient.

Since

$$\frac{d^2 V}{db_0^2} = 2S_x^2 > 0,$$

the variance is minimized at this value.

Minimum Variance Result

At

$$b_0 = \frac{S_{xy}}{S_x^2},$$

the minimum variance becomes

$$V_{\min}(\bar{y}_{lr}) \approx \left(1 - \frac{n}{N}\right) \frac{S_y^2(1 - \rho^2)}{n}$$

where

$$\rho = \frac{S_{xy}}{S_x S_y}$$

is the population correlation coefficient between X and Y .

Important Result

As the correlation $|\rho|$ between X and Y increases, the variance of the regression estimator decreases.

SRS, Ratio and Regression Estimation

Estimator	Basic Form	Main Idea
Sample Mean	$\bar{y}$	Uses only the study variable.
Ratio	$\bar{y} \left(\frac{\bar{X}}{\bar{x}} \right)$	Uses the auxiliary variable through a ratio.
Regression	$\bar{y} + b(\bar{X} - \bar{x})$	Uses the linear relationship between X and Y .

Rule of Thumb

Regression estimation is especially useful when X and Y have a strong approximately linear relationship.

Advantages of Regression Estimation

- ① Makes use of additional information from an auxiliary variable.
- ② Can provide greater precision than the sample mean.
- ③ Particularly useful when X and Y are strongly correlated.
- ④ Does not require the regression line to pass through the origin.
- ⑤ Provides a natural adjustment when the sample mean $\bar{x}$ differs from the known population mean $\bar{X}$.

Condition for Effective Use

The auxiliary variable should be meaningfully related to the study variable and its population mean should be known or reliably available.

Regression Estimation in Stratified Sampling

Suppose the population is divided into L strata. For stratum h :

N_h = population size, n_h = sample size.

Let

$$W_h = \frac{N_h}{N}$$

be the stratum weight.

The regression estimator can be constructed separately within each stratum.

Separate Regression Estimator

$$\hat{\bar{Y}}_{reg,sep} = \sum_{h=1}^L W_h [\bar{y}_h + b_h(\bar{X}_h - \bar{x}_h)]$$

Separate Regression Estimator

For each stratum h ,

$$\hat{\bar{Y}}_{reg,h} = \bar{y}_h + b_h(\bar{X}_h - \bar{x}_h).$$

Therefore,

$$\hat{\bar{Y}}_{reg,sep} = \sum_{h=1}^L W_h \hat{\bar{Y}}_{reg,h}$$

or

$$\hat{\bar{Y}}_{reg,sep} = \sum_{h=1}^L W_h [\bar{y}_h + b_h(\bar{X}_h - \bar{x}_h)] .$$

- Each stratum has its own regression coefficient b_h .
- Useful when relationships between X and Y differ across strata.

Variance of Separate Regression Estimator

The variance is obtained by combining the variance contributions from all strata.

$$V(\hat{Y}_{reg,sep}) = \sum_{h=1}^L W_h^2 V(\hat{Y}_{reg,h})$$

Under simple random sampling within each stratum,

$$V(\hat{Y}_{reg,h}) \approx \left(1 - \frac{n_h}{N_h}\right) \frac{S_{yh}^2(1 - \rho_h^2)}{n_h}.$$

Hence,

$$V(\hat{Y}_{reg,sep}) \approx \sum_{h=1}^L W_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{S_{yh}^2(1 - \rho_h^2)}{n_h}.$$

Combined Regression Estimator

Instead of estimating a separate regression coefficient in every stratum, a common regression coefficient can be used.

The combined estimator can be written as

$$\hat{\bar{Y}}_{reg,com} = \bar{y}_{st} + b(\bar{X} - \bar{x}_{st})$$

where

$$\bar{y}_{st} = \sum_{h=1}^L W_h \bar{y}_h$$

and

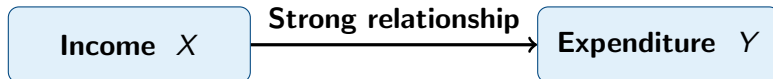
$$\bar{x}_{st} = \sum_{h=1}^L W_h \bar{x}_h.$$

Separate vs. Combined Regression Estimator

	Separate	Combined
Regression coefficient	b_h for each stratum	One common b
Estimator	$\sum W_h[\bar{y}_h + b_h(\bar{X}_h - \bar{x}_h)]$	$\bar{y}_{st} + b(\bar{X} - \bar{x}_{st})$
Useful when	Relationships differ substantially	A common relationship is reasonable

Example: Regression Estimation in Practice

Suppose a researcher wants to estimate the average annual household expenditure Y . An auxiliary variable is household income X .



If the population mean income $\bar{X}$ is known from an administrative source, it can be used to improve estimation of the population mean expenditure.

$$\hat{\bar{Y}}_{reg} = \bar{y} + b(\bar{X} - \bar{x})$$

Key Takeaways

- ➊ Regression estimation uses an auxiliary variable correlated with the study variable.
- ➋ The population mean $\bar{X}$ of the auxiliary variable is assumed to be known.
- ➌ The basic regression estimator is

$$\bar{y}_{lr} = \bar{y} + b(\bar{X} - \bar{x}).$$

- ➍ A strong correlation between X and Y can lead to substantial gains in precision.
- ➎ The minimum variance occurs at the population regression coefficient.
- ➏ Regression estimation can also be applied within stratified sampling.

Quick Check

Question 1

Why is an auxiliary variable used in regression estimation?

Question 2

What happens to the regression estimator when $\bar{x} = \bar{X}$?

Question 3

What happens to the variance when the correlation between X and Y becomes stronger?

Think

Under what circumstances would you prefer regression estimation over ratio estimation?

Thank You

Questions and Discussion

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